

ECE Probabilistic Systems Analysis

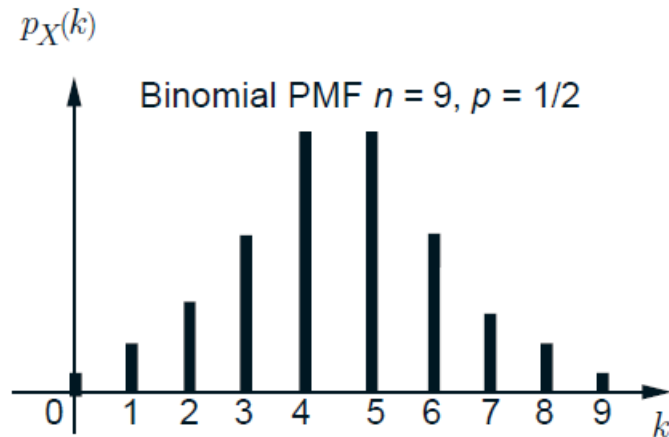
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Continuous Random Variables

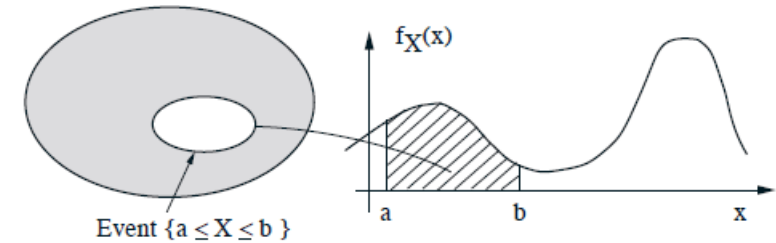
Probability Mass Function or (PMF) $p_X(x)$



$$\mathbf{P}(a \leq X \leq b) = \sum_{x: a \leq x \leq b} p_X(x)$$

$$p_X(x) \geq 0 \quad \sum_x p_X(x) = 1$$

Probability Density Function or (PDF) $f_X(x)$

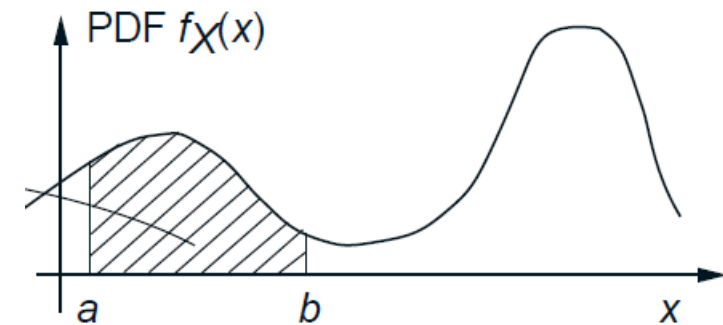


The probability that the value of X falls within an interval $[a, b]$ is:

$$\mathbf{P}(a \leq X \leq b) = \int_a^b f_X(x) dx,$$

$$f_X(x) \geq 0 \quad \int_{-\infty}^{\infty} f_X(x) dx = 1$$

$f_X(x)$ is the “probability mass per unit length” near x



$$\mathbf{P}(a \leq X \leq a + \delta) \approx f_X(a) \cdot \delta$$

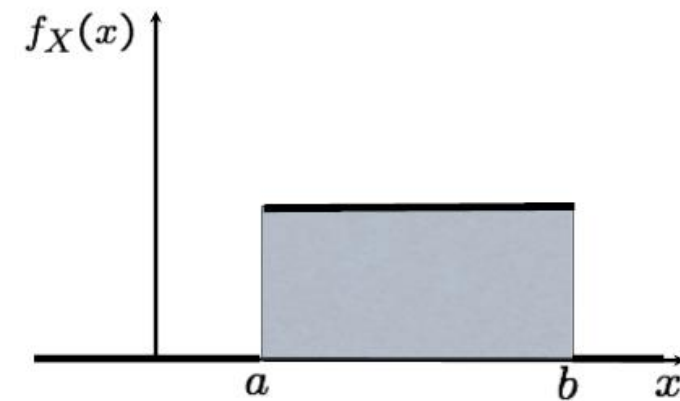
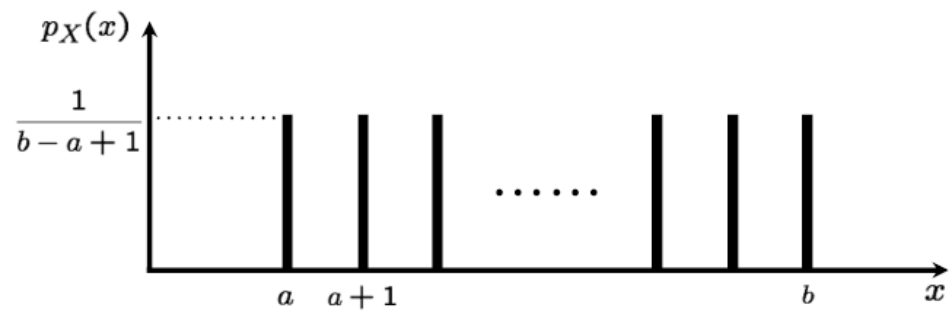
$$\mathbf{P}(X = a) = 0$$

$$\mathbf{P}(X = a) = \int_a^a f_X(x) dx = 0$$

For this reason, including or excluding the endpoints of an interval has no effect on its probability

$$\mathbf{P}(a \leq X \leq b) = \mathbf{P}(a < X < b) = \mathbf{P}(a \leq X < b) = \mathbf{P}(a < X \leq b)$$

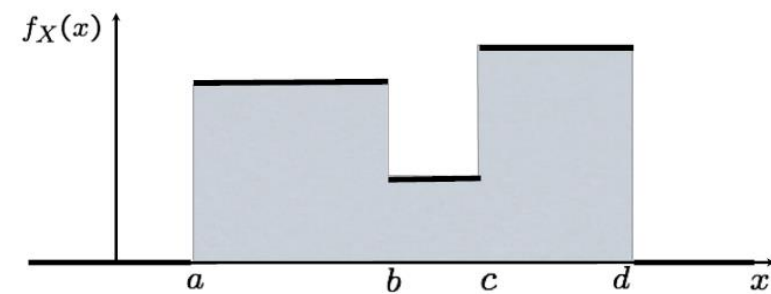
Continuous Uniform Random Variable



$$1 = \int_a^b c \, dx = c \int_a^b dx = c(b-a),$$

$$c = \frac{1}{b-a}.$$

$$f_X(x) = \begin{cases} c & \text{if } a \leq x \leq b, \\ 0 & \text{otherwise,} \end{cases}$$

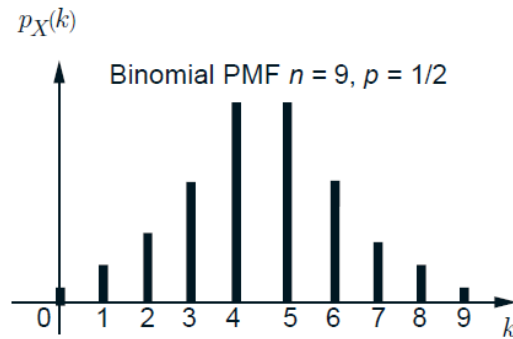


Alvin's driving time to work is between 15 and 20 minutes if the day is sunny, and between 20 and 25 minutes if the day is rainy, with all times being equally likely in each case. Assume that a day is sunny with probability $2/3$ and rainy with probability $1/3$. What is the PDF of the driving time, viewed as a random variable X ?

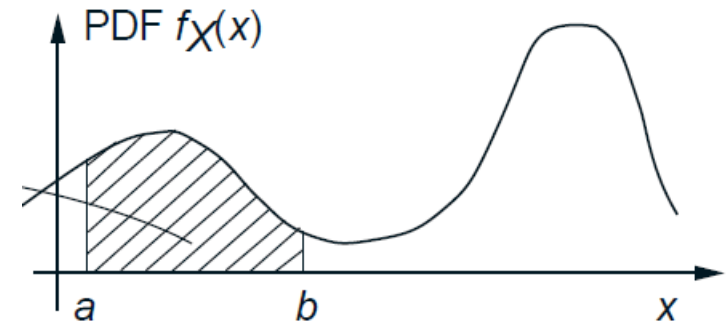
$$f_X(x) = \begin{cases} c_1 & \text{if } 15 \leq x < 20, \\ c_2 & \text{if } 20 \leq x \leq 25, \\ 0 & \text{otherwise,} \end{cases}$$

$$c_1 = \frac{2}{15}, \quad c_2 = \frac{1}{15}.$$

Expectation/ mean of a Continuous Random Variable



$$\mathbf{E}[X] = \sum_x x p_X(x)$$



$$\mathbf{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx.$$

$\mathbf{E}[X]$ can be interpreted as the “center of gravity” of the probability law and, also, as the anticipated average value of X in a large number of independent repetitions of the experiment.

Mean and Variance

The expectation of X is defined by

$$\mathbf{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx.$$

The variance of X is defined by

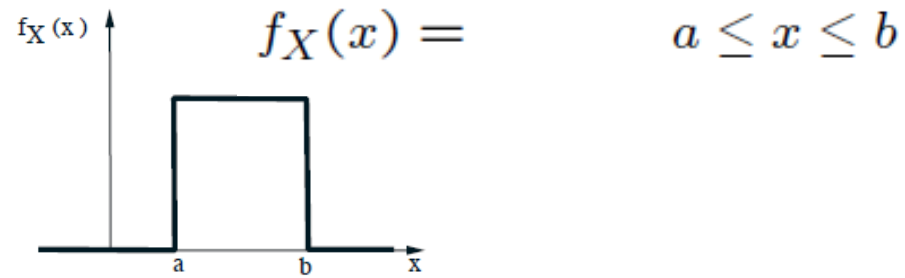
$$\text{var}(X) = \mathbf{E}[(X - \mathbf{E}[X])^2] = \int_{-\infty}^{\infty} (x - \mathbf{E}[X])^2 f_X(x) dx$$

$$\text{var}(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2.$$

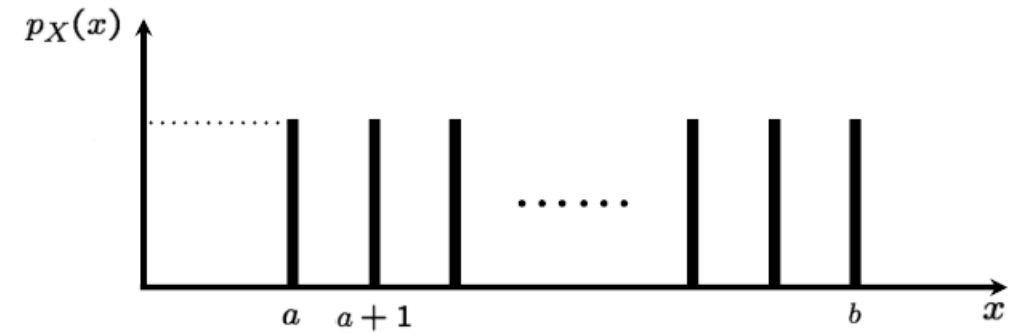
Standard Deviation

$$\sigma(x) = \sqrt{\text{var}(x)}$$

Mean and Variance of the Uniform Random Variable



$$E[X] =$$



$$E[X] = \frac{a + b}{2}$$

$$\text{var}(X) = \frac{1}{12}(b - a)(b - a + 2)$$

$$E[X] = \frac{a + b}{2}$$

Mean and Variance of the Uniform Random Variable

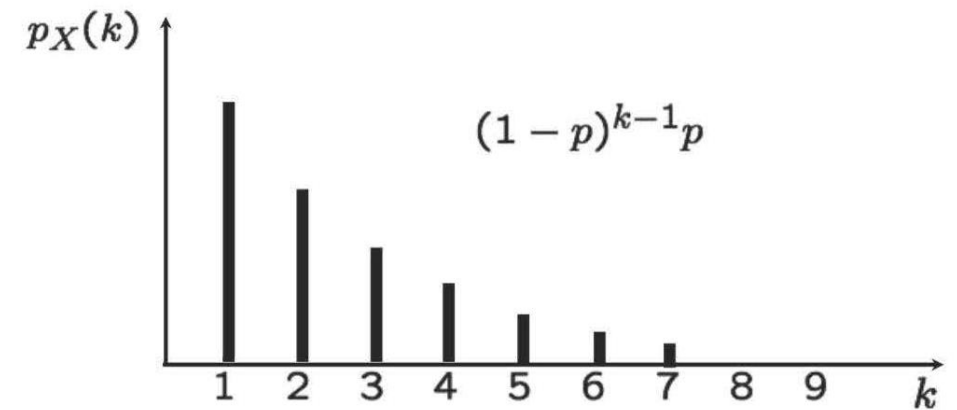
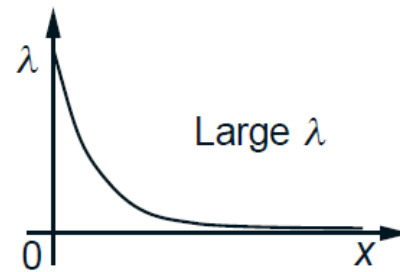
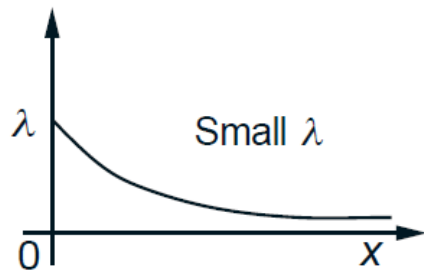
$$\mathbf{E}[X^2] =$$

$$\text{var}(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2 = \frac{a^2 + ab + b^2}{3} - \frac{(a + b)^2}{4} = \frac{(b - a)^2}{12},$$

$$\sigma_X^2 = \int_a^b \left(x - \frac{a + b}{2}\right)^2 \frac{1}{b - a} dx = \frac{(b - a)^2}{12}$$

Mean and Variance of the Exponential Random Variable

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$



$$\mathbf{E}[X] = \frac{1}{\lambda}.$$

$$\mathbf{E}[X^2] = \frac{2}{\lambda^2}.$$

$$\mathbf{var}(X) = \text{var}(X) = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}.$$

CUMULATIVE DISTRIBUTION FUNCTIONS or CDF

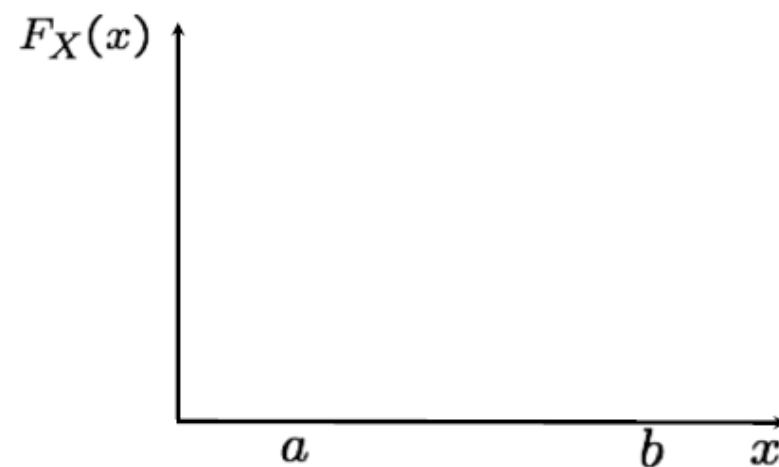
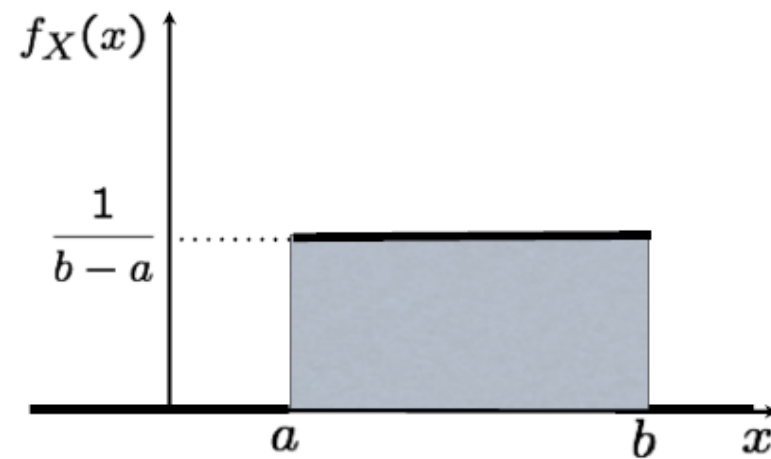
CDF $F_X(x)$ “accumulates” probability “up to” the value x

$$F_X(x) = \mathbf{P}(X \leq x) = \begin{cases} \sum_{k \leq x} p_X(k) & X: \text{discrete,} \\ \int_{-\infty}^x f_X(t) dt & X: \text{continuous.} \end{cases}$$

Continuous Random Variables

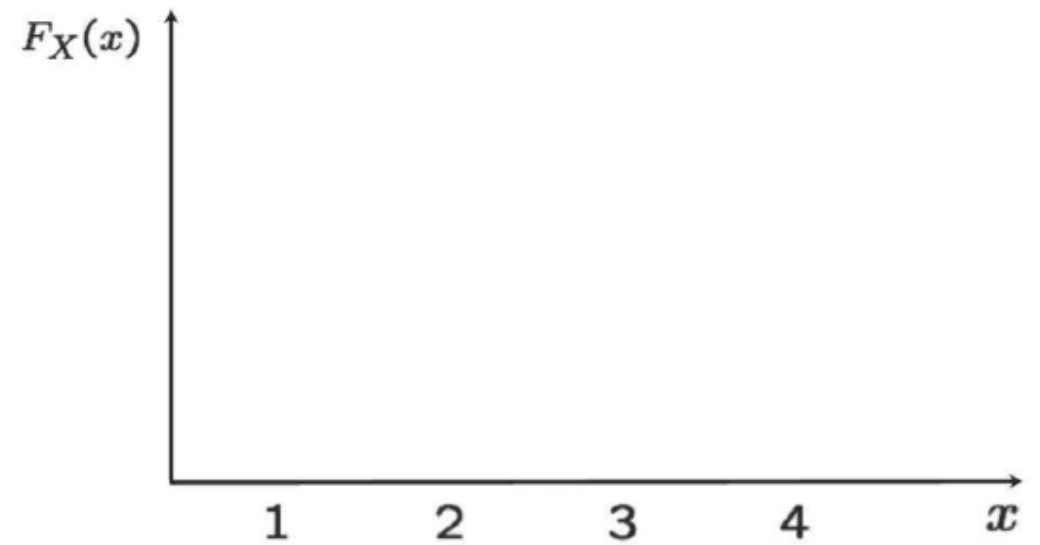
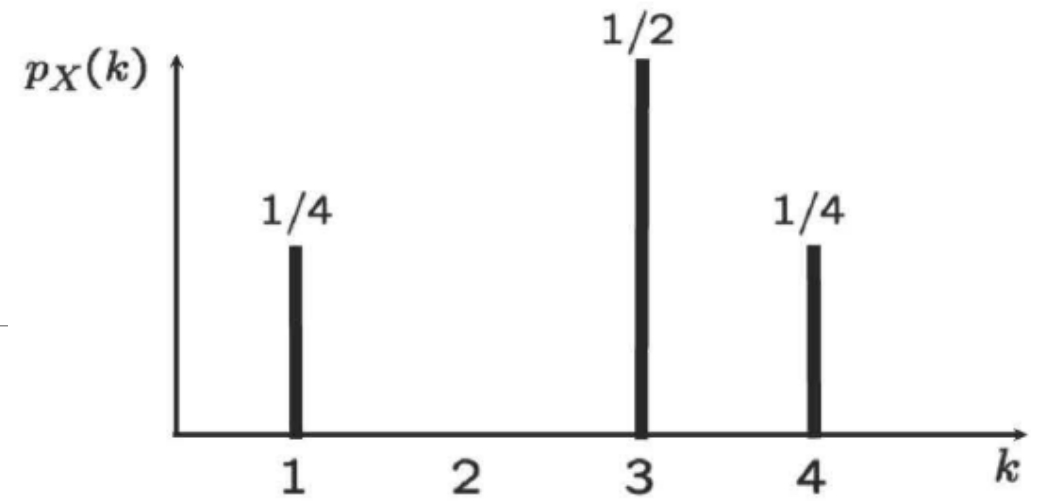
$$F_X(x) = \int_{-\infty}^x f_X(t) dt,$$

$$f_X(x) = \frac{dF_X}{dx}(x).$$



Discrete Random Variables

$$F_X(x) = \mathbf{P}(X \leq x) = \sum_{k \leq x} p_X(k)$$



Properties of a CDF

- F_X is monotonically nondecreasing:

$$\text{if } x \leq y, \text{ then } F_X(x) \leq F_X(y).$$

- $F_X(x)$ tends to 0 as $x \rightarrow -\infty$, and to 1 as $x \rightarrow \infty$.
- If X is discrete, then F_X has a piecewise constant and staircase-like form.
- If X is continuous, then F_X has a continuously varying form.

Let X be a continuous random variable with the following PDF

$$f_X(x) = \begin{cases} ce^{-x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

where c is a positive constant.

- a. Find c .
- b. Find the CDF of X , $F_X(x)$.
- c. Find $P(1 < X < 3)$.

$$\begin{aligned} 1 &= \int_{-\infty}^{\infty} f_X(u) du \\ &= \int_0^{\infty} ce^{-u} du \\ &= c \left[-e^{-u} \right]_0^{\infty} \\ &= c. \end{aligned} \qquad F_X(x) = \begin{cases} 1 - e^{-x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

c. We can find $P(1 < X < 3)$ using either the CDF or the PDF. If we use the CDF, we have

$$P(1 < X < 3) = F_X(3) - F_X(1) = [1 - e^{-3}] - [1 - e^{-1}] = e^{-1} - e^{-3}.$$

Equivalently, we can use the PDF. We have

$$\begin{aligned} P(1 < X < 3) &= \int_1^3 f_X(t) dt \\ &= \int_1^3 e^{-t} dt \\ &= e^{-1} - e^{-3}. \end{aligned}$$

Let X be a discrete random variable with range $R_X = \{1, 2, 3, \dots\}$. Suppose the PMF of X is given by

$$P_X(k) = \frac{1}{2^k} \text{ for } k = 1, 2, 3, \dots$$

- a. Find and plot the CDF of X , $F_X(x)$.
- b. Find $P(2 < X \leq 5)$.
- c. Find $P(X > 4)$.

First, note that this is a valid PMF. In particular,

$$\sum_{k=1}^{\infty} P_X(k) = \sum_{k=1}^{\infty} \frac{1}{2^k} = 1 \text{ (geometric sum)}$$

- a. To find the CDF, note that

$$\text{For } x < 1, \quad F_X(x) = 0.$$

$$\text{For } 1 \leq x < 2, \quad F_X(x) = P_X(1) = \frac{1}{2}.$$

For $2 \leq x < 3$, $F_X(x) = P_X(1) + P_X(2) = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$.

In general we have

For $0 < k \leq x < k+1$,

$$F_X(x) = P_X(1) + P_X(2) + \dots + P_X(k)$$

$$= \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^k} = \frac{2^k - 1}{2^k}.$$

b. To find $P(2 < X \leq 5)$, we can write

$$P(2 < X \leq 5) = F_X(5) - F_X(2) = \frac{31}{32} - \frac{3}{4} = \frac{7}{32}.$$

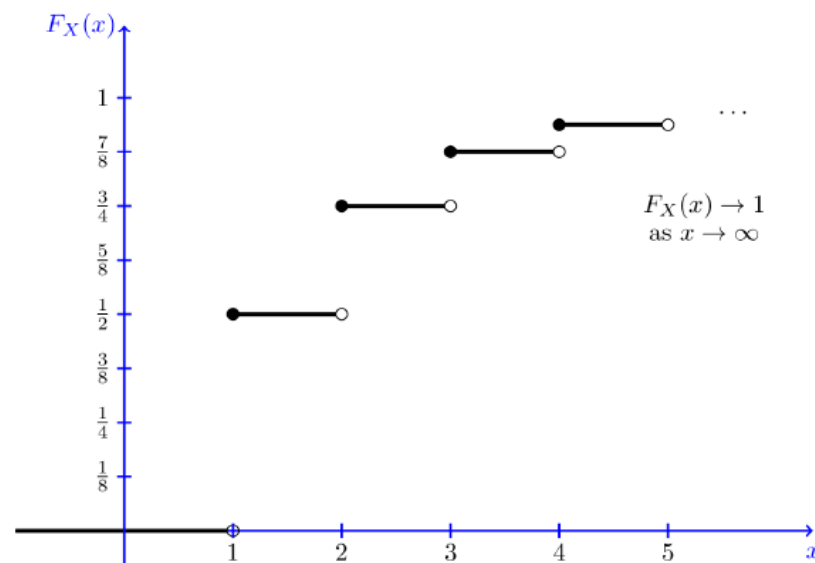
Or equivalently, we can write

$$P(2 < X \leq 5) = P_X(3) + P_X(4) + P_X(5) = \frac{1}{8} + \frac{1}{16} + \frac{1}{32} = \frac{7}{32},$$

which gives the same answer.

c. To find $P(X > 4)$, we can write

$$P(X > 4) = 1 - P(X \leq 4) = 1 - F_X(4) = 1 - \frac{15}{16} = \frac{1}{16}.$$



1. Let Z be a continuous random variable with probability density function

$$f_z(z) = \begin{cases} \gamma(1 + z^2), & \text{if } -2 < z < 1, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) For what value of γ is this possible?
- (b) Find the cumulative distribution function of Z .

1. (a) We know that the PDF must integrate to 1. Therefore we have

$$\int_{-\infty}^{\infty} f_Z(z) dz = \int_{-2}^1 \gamma(1+z^2) = \gamma \left(z + \frac{1}{3}z^3 \right) \Big|_{-2}^1 = 6\gamma.$$

From this we conclude $\gamma = 1/6$.

- (b) To find the CDF, we integrate:

$$\begin{aligned} F_Z(z) &= \int_{-\infty}^z f_Z(t) dt = \begin{cases} 0, & \text{if } z < -2, \\ \frac{1}{6} \left(t + \frac{1}{3}t^3 \right) \Big|_{-2}^z, & \text{if } -2 \leq z \leq 1, \\ 1, & \text{if } z > 1 \end{cases} \\ &= \begin{cases} 0, & \text{if } z < -2, \\ \frac{1}{6} \left(z + \frac{1}{3}z^3 + \frac{14}{3} \right), & \text{if } -2 \leq z \leq 1, \\ 1, & \text{if } z > 1. \end{cases} \end{aligned}$$